

**Title: Impact of Cancer-Associated Mutations on DNA Sequence Composition of different species
(human,rat,mouse,mutant)**

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1. Introduction

Understanding genetic similarity and sequence composition is fundamental in molecular biology and bioinformatics. Phylogenetic analysis helps infer evolutionary relationships between sequences, while nucleotide composition metrics such as GC content and motif frequency provide insight into structural and functional properties of DNA.

The purpose of this study is to combine phylogenetic tree construction with sequence-level statistical analysis in a unified workflow, enabling both evolutionary and compositional insights from a single dataset.

2. Dataset Description

The dataset consists of four short DNA sequences labeled **Human, Mouse, Rat, and Mutant**. These sequences are highly similar, differing only by a few nucleotides, making them ideal for comparative and phylogenetic analysis.

Each sequence includes:

- Identifier (species/variant name)

- DNA nucleotide sequence
 - Derived variables such as length, GC content, motif count, and translated protein
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3. FASTA Sequences

The sequences used in this study are presented below in standard FASTA format:

>Human

ATGGTGAAGGTCTGAGGAGGAAGTCTTCTC

>Mouse

ATGGTGAAGGTCTGAGGAGGAAGTCTTCTT

>Rat

ATGGTGAAGGTCTGAGGAGGAAGTCTTCCT

>Mutant

ATGGTGAAGGTCTGAGGTGGAAGTCTACCT

These sequences represent short gene fragments with minor variations. The Mutant sequence contains substitutions that differentiate it from the others.

4. Methodology

The analysis was performed using Python with Biopython, Pandas, and Matplotlib.

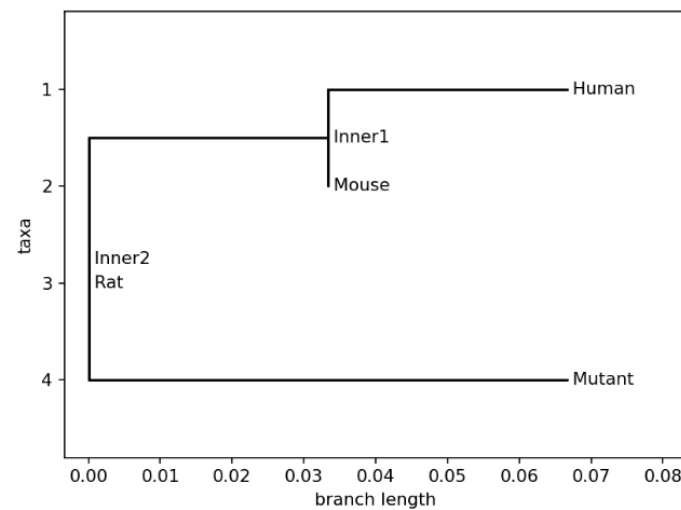
Steps included:

1. Sequence definition and formatting (FASTA structure)
2. Multiple sequence alignment
3. Distance matrix computation using identity scoring
4. Phylogenetic tree construction via Neighbor-Joining
5. Feature extraction:
 - Sequence length
 - GC content
 - Motif ("ATG") count

- Protein translation
6. Data organization into a dataframe
 7. Visualization of results through charts
-

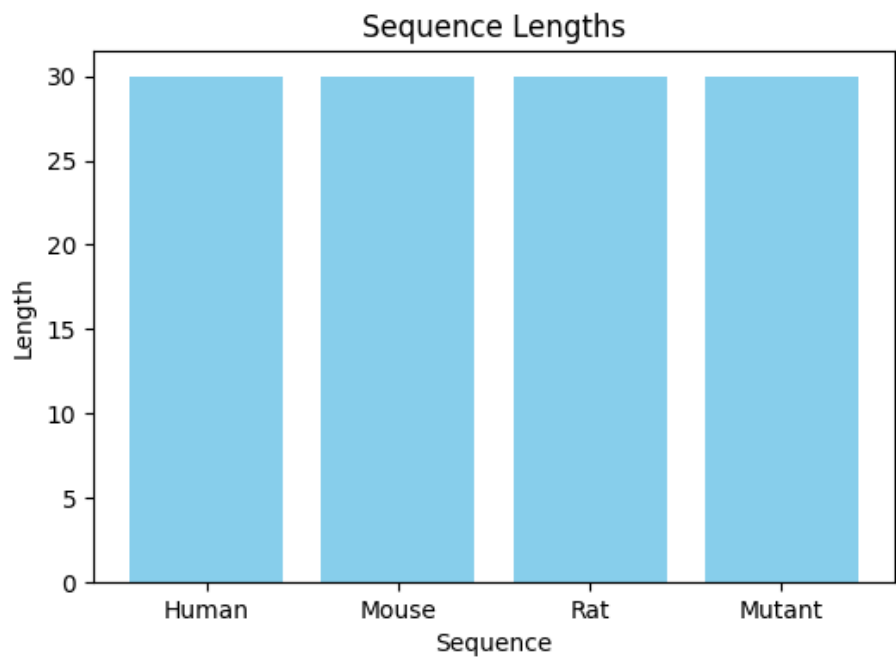
5. Analysis

Phylogenetic Tree



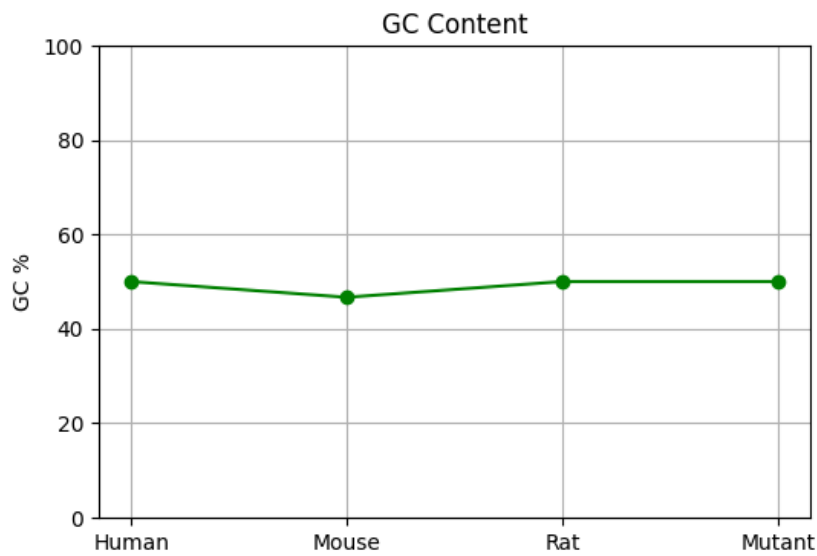
The tree shows close clustering among Human, Mouse, and Rat, with the Mutant slightly diverging.

Sequence Length Comparison



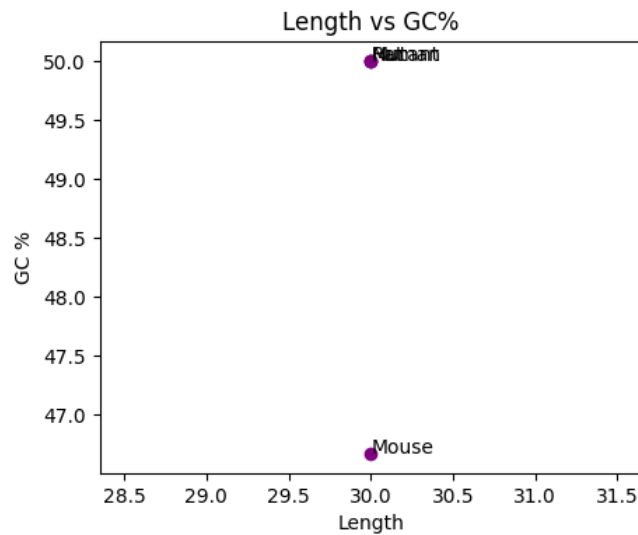
All sequences have nearly identical lengths, indicating consistency in dataset structure.

GC Content Trend



GC content is highly similar across all sequences, with only slight variation in the Mutant.

Length vs GC Scatter



The clustering confirms uniform composition across sequences.

6. Interpretation

The phylogenetic tree indicates strong conservation among Human, Mouse, and Rat sequences, suggesting evolutionary similarity. The Mutant sequence shows slight divergence due to nucleotide substitutions.

Consistent GC content implies similar structural stability and thermodynamic properties across sequences. Since GC-rich regions are more stable, this uniformity suggests comparable biological behavior.

The presence of the "ATG" start codon in all sequences indicates that they are likely functional and capable of translation. However, small sequence differences may result in slight variations in protein products, potentially affecting function.

Overall, the analysis demonstrates how small mutations can be detected and interpreted using combined phylogenetic and statistical approaches.

7. Output / Results Summary

This study successfully integrates phylogenetic and sequence composition analysis within a single framework. The results show high similarity among sequences, with only minor divergence in the Mutant variant.

The phylogenetic tree clearly visualizes evolutionary relationships, while statistical metrics confirm compositional consistency. Visualizations enhance interpretation by making patterns more accessible.

The findings highlight that even small datasets can yield meaningful biological insights when analyzed systematically and supported by clear visual representations.
